

Satya Sri Lasya Mundru

913-280-0877 | lasyamundru11@gmail.com | [LinkedIn](#)

Professional Summary

Results-oriented **Backend Software Engineer** with **3.5 years of product development experience** designing and scaling **distributed backend systems** in Banking and Healthcare domains. Proficient in **Python development, AI/ML model integration, data engineering, and cloud-based solutions**. Skilled in **Python, SQL, C++, Golang, Java, TensorFlow, PyTorch, Flask, FastAPI, AWS, Azure, Docker, Kubernetes, and CI/CD pipelines**. Proven track record of delivering **enterprise-scale APIs, ETL workflows, AI/ML deployments, and data pipelines** that support clients like **Birlasoft, Elevance Health, and Fifth Third Bank**. Skilled in **AWS, Azure, and GCP**, with a strong focus on **performance optimization, observability, and end-to-end lifecycle ownership** from design to production in Agile environments.

Technical Skills

- **Programming & Scripting:** Python, Java, SQL, Shell Scripting, C++, Golang (familiar)
 - **AI/ML Frameworks:** TensorFlow, PyTorch, Scikit-learn, Hugging Face, NLTK
 - **Web & API Frameworks:** Flask, FastAPI, Django, REST, GraphQL
 - **Cloud Platforms:** AWS (EC2, S3, Lambda, SageMaker, Glue, Redshift), Azure (Data Lake, AKS, Data Factory), GCP (familiar)
 - **Systems Design:** Microservices Architecture, Distributed Systems, Data-intensive Systems, Event-driven Architecture, Fault Tolerance
 - **Data Engineering:** Apache Spark, PySpark, Pandas, NumPy, Airflow, dbt, Kafka
 - **Databases:** MySQL, PostgreSQL, MongoDB, SQL Server, Snowflake
 - **DevOps & CI/CD:** Jenkins, GitHub Actions, Docker, Kubernetes, Terraform, Ansible
 - **Monitoring & Logging:** CloudWatch, Splunk, ELK, Prometheus, Grafana
 - **Visualization & Reporting:** Tableau, Power BI, Matplotlib, Seaborn
-

Professional Experience

Project: Intelligent Fraud Detection Platform (Banking Domain)

Python Developer | Fifth Third Bank, Kentwood, MI

Jan 2025 – Present

- Designed and deployed **AI/ML pipelines** for fraud detection using **PyTorch** and **Scikit-learn**, reducing false positives by 20%.
- Developed **FastAPI microservices** for real-time transaction validation integrated across multiple AWS regions.
- Automated data ingestion and ETL workflows using **Airflow** and **AWS Redshift**, optimizing throughput
- Built **Power BI dashboards** for portfolio risk monitoring and loan default predictions.
- Leveraged **AWS Lambda + S3** for event-driven architectures, cutting data latency by 40%.
- Led end-to-end feature ownership from architecture to production rollout in a highly autonomous environment.
- Implemented **schema validation & data governance** with **Great Expectations**.
- Enhanced observability through **Prometheus, Grafana, and ELK**, reducing debugging time by 35%.
- Partnered with data science teams for model training & deployment pipelines.
- Participated in Agile ceremonies, delivering sprint-based modules.
- Collaborated with cross-functional teams across **data science, DevOps, and product** to align platform goals

Environment: Python, FastAPI, PyTorch, Scikit-learn, SQL Server, AWS (Redshift, Lambda, S3, EC2), Airflow, Jenkins, Docker, Kubernetes, Tableau, Power BI, ELK, Golang.

Project: AI-Driven Claims Analytics (Healthcare Domain)

AI Python Developer | Elevance Health, Atlanta, GA

Jun 2024 – Dec 2024

- Designed **NLP pipelines** with **Hugging Face Transformers** and **NLTK** for claims text classification.
- Built **Python ETL frameworks** to process large-scale claims into **Azure Data Lake**.
- Deployed **predictive models** in **TensorFlow** for patient readmission analysis.
- Integrated **FastAPI endpoints** for real-time claim adjudication, reducing processing time by 30%.
- Orchestrated **Azure Data Factory pipelines** for HIPAA-compliant ETL workflows.
- Processed multi-terabyte healthcare datasets with **PySpark**, reducing query runtimes.
- Deployed AI models with **Docker** on **Azure Kubernetes Service (AKS)** for scalability.
- Implemented **Prometheus/Grafana monitoring** for pipeline and model observability.
- Ensured compliance by validating data with **Great Expectations**.
- Developed **Power BI dashboards** for claim cost, fraud detection, and provider efficiency.
- Worked with compliance officers and data scientists to align AI solutions with regulatory needs.

Environment: Python, TensorFlow, Hugging Face, PySpark, Azure Data Lake, Azure Data Factory, AKS, Docker, Prometheus, Grafana, Power BI.

Project: Digital Banking Automation Suite (Banking Domain)

Python Developer | Birlasoft, Hyderabad, India

Jun 2021 – Jul 2023

- Developed **Python automation frameworks** for loan origination & transaction processing.
- Created **Flask REST APIs** to serve banking microservices.
- Optimized **SQL queries** and **stored procedures** for financial transactions.
- Migrated **legacy ETL jobs** into **PySpark pipelines**, improving runtime by 40%.
- Automated **regulatory compliance reports** with Python + Tableau/Power BI.
- Built ML-based models using **Scikit-learn** to predict loan approvals.
- Implemented **CI/CD pipelines** in Jenkins for automation of deployments.
- Deployed **containerized applications** on **Docker & Kubernetes**.
- Enhanced fraud monitoring with **anomaly detection algorithms**.
- Designed scalable **data ingestion pipelines** with Pandas & NumPy.
- Participated in Agile sprints, ensuring delivery of high-quality modules.

Environment: Python, Flask, Pandas, NumPy, Scikit-learn, SQL, PySpark, Jenkins, Docker, Kubernetes, Tableau, Power BI.

Education

- Master of Science in Computer Science (2023 – 2025), University of Central Missouri, USA
- Bachelor of Technology in Computer Science (2019 – 2023), Andhra University, India

Certifications

- AWS Certified Developer – Associate
 - Microsoft Azure Fundamentals (AZ-900)
 - TensorFlow Developer Certificate
 - SQL for Data Science – Coursera
-